

JustInTime

FloatFry's Manufacturing Resource Planning

BACKGROUND

The Problems

FloatFry is a non-stick cookware manufacturer operating in a unique way. Since the start of the business in mid-1993, the company has always been developing products which are exclusive and family-oriented, with a nice touch of personalization. They are very proud of their products and only sell them to the highest-end market. All their cookware is claimed to be oven-safe up to 280⁰, dishwasher-safe, and pressed from extremely durable Titanium giving them scratch-resistant, ultra-slick surface features. Additionally, FloatFry can take special orders from their customers to put their signature on the products, which has catapulted FloatFry's cookware to be one of the poshest products in the market.



Image Source: [The Independent](#)

FloatFry do not mass-produce their cookware, they make them by order. FloatFry was built and is run closely by a family, the Floats, who started the business in a 3x2 meter granny flat. They used to order materials only from local businesses and build their products manually. They would have never dreamt of FloatFry being so popular that they are now swamped by hundreds of orders every day. Problems start to occur when their local partners can no longer keep up with material demands, unexpected price changes impact the production costs more significantly with more products being produced, and sales forecast seems to be less and less reliable.

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The Vision

In early 2022, FloatFry initiated a manufacturing process redesign as part of the company's commitment to its business process overhaul. The redesign work is highly critical to the running of the business and the company must take precautionary measures for every change made to accommodate the newly proposed manufacturing information systems. However, at the same time, they need to make sure that the change process should not be dragged on too long and that it may unsettle the running of the business. The management team has projected that it will take around 2-3 years for the redesigned business processes to run effectively. The new manufacturing information system, JustInTime, therefore, needs to launch by the first quarter of 2025.

FloatFry, however, never think of JustInTme as a new way to do their business. Rather, they want the new digital system to be the manifestation of their long-guarded business vision and processes they are really proud of. The system will operate across multiple business areas, starting from supply chain management, product definition across the product lifecycle, and resource management (including machines, materials, and people) to schedule, order execution and production analysis. JustInTime will capture data directly from physical manufacturing equipment, monitor and manage resources – including materials and suppliers, and provide services to other business processes, all in real-time.

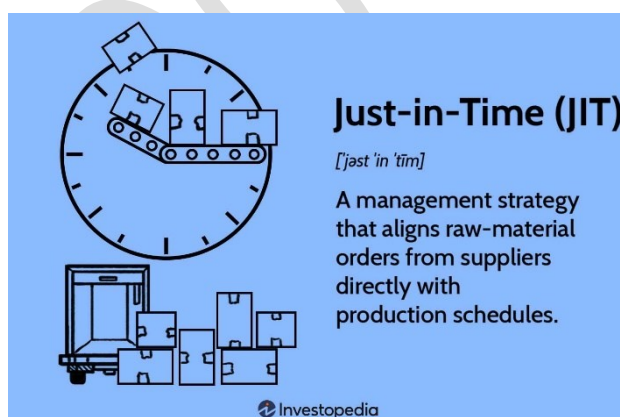


Image Source: [MBA Skool](#)

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REQUIREMENT COLLECTION

The following are some interviews conducted during the inception phase. The requirement collection has involved more stakeholders and potential system users. However, for the sake of clarity and simplicity, only some of them are presented below.

INTERVIEW 01: Mattie Float – Stamping Machine Operator at FloatFry.

Q	<i>Good morning. Could you please start by introducing yourself?</i>
A	Good morning to you too. My name is Mattie, and I am a Stamping Machine Operator, or Stamper we call ourselves, at FloatFry. I am also one of the founders of FloatFry.
Q	<i>Thanks, Mattie. Could you tell me about your job and what you do daily?</i>
A	Mind you, in the early days, everyone kind of like did everything because there were not many of us, and the business was far less complicated. However, we have been trying to be more organized and to delegate jobs to new employees joining us lately. I am responsible for operating all the machines in this plant, mainly our big friends here, the stamping machines. Mind you, there are eight Stampers here, that is because we have got hundreds of machines to run in this plant. The machines are of different types and can be programmed to do specific jobs, such as conveying components, assembling bases, lids or handles, and pressing. For finishing jobs, we have coating machines, plotter printing, cutting and heat-seal joining machines, among others. Some machines are complicated and require delicate treatments. Those coating machines there, for instance, have a moving coating tank and mechanical or pneumatic components – depending on the model – that control the angle and pressure of the tank. Frequent calibration and maintenance is the key to ensuring the results have high precision. We, the Stampers, are the only ones with the skills and, therefore, authorized to operate these machines.

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	Stampers possess different skill sets from each other. I mean, I can run almost all the machines in this plant, but I am only certified to set and operate the big stamping machines, conveyor belts, plotter printers, cutting and heat-seal joining machines, and several assembly machines. Well, on average, each Stamper has certifications for 3 to 4 different machines. That is why the scheduling of Stamper and machines is very crucial here.
Q	<i>Can you guide me through your activities on a typical day from the beginning to the end of the business day?</i>
A	Well, we have 2 working shifts for Stampers here, 8 hours long each. I usually start my shift by clocking in and then studying my job sheet. My job sheet is prepared by the Floor Manager at least one day before. On the sheet is information about which machines I will be operating, from and to what time, and the job cards. A job card contains a unique identifier and information about what task I need to perform on which machine, at what time, applying which design pattern, and using what materials. I then go to the machine mentioned in my first job card and start setting it up. I don't need to prepare the materials; the POs - Production Operatives do that for me. I mean, I know how to handle materials, but to choose the right materials with the right specifications for a design, we have got experts to do that. Then I check the design on the machine's computer if it has been sent through the network, set the machine up to the design specifications, and start the production. When I finish, I move to the next job card right away because another stamper may use the machine for different jobs.
Q	<i>You mentioned the network, are all machines here connected to the network?</i>
A	Yes, of course. Every machine has its own mechanical parts that do the job, an electronic control terminal that controls how the machine works, and a digital workstation that works as an interface between the Stamper and the control terminal. The workstation is connected to the network and can exchange data in real-

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	time, which means it can be automated to some extent. However, the setting up of the mechanical parts of the machine, such as spray needles, cutting blades, et cetera, and the setting up of the materials, still require manual work. That is why Stampers are very important to the running of this plant.
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INTERVIEW 02: Abby Float - Floor Manager at FloatFry

Q	<i>Hi. Could you please tell me more about your responsibilities?</i>
A	Yes, of course. I am a Floor Manager here at FloatFry, which means I am responsible for the running of this plant. By running, I am referring to the machines and the operators. I mean, I am not responsible for the production and the production schedule, the Production team is, but I do provide them with information about the availability of machines and Stampers. Another part of my job is to maintain the machines, making sure they are ready when needed. I am also responsible for implementing the production line. I mean, usually, the production team, or the Prod, lets me know in a meeting that they are preparing to manufacture a new item, for instance, and they provide me with the product specifications and technical details. We then discuss what manufacturing processes are required, what machines may be involved, and how the process will flow. I then make sure that the production line is implemented and ready, and that the machines involved are set up properly. But this sort of thing does not come regularly, only when we have a new product to release.
Q	<i>You mentioned feeding information to the Production team, could you elaborate?</i>
A	I have all the data about machines, operators, and the timetable, which is about which Stamper is using which machine. Each machine can do one or more jobs every cycle, either simultaneously or sequentially, and has maximum production capacity per second, minute, hour, or daily. Although the machines are connected to the network, every one of them still requires a Stamper to set up correctly before it can do the job. Nevertheless, machine data, such as what job a machine is performing, when it starts, when it ends, the target capacity, the cycle speed, any detected

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	troubles, and troubleshooting or maintenance required, among others, are available in real-time. Now, you must understand that when I said a machine needs an operator, it is not just any operator. Only Stampers who have had extensive training and earned a certificate on a machine can operate the machine. We are proud to say that all our Stampers have more than one certification, so we are never short of skilled resources. The problem now is how to generate a timetable of what item needs to be produced, what types of jobs are involved in the manufacturing processes, what machine or machines are required to do the jobs, how the production line is going to look, who are going to operate the machines, as well as when and where. So, I usually bring this kind of information to the meeting with the production team, and we make sure that everything is set up properly before we start production.
Q	<i>How can the newly proposed Manufacturing Resources Planning help you with your job?</i>
A	Well, I guess I would be happy to have a system that can help me generate and manage the timetable. Mind you, although the machines are connected to the network, each machine is still using the manufacturer's proprietary software that came with it. So, you can say that I still create the timetable manually. It is only because I have been working here for years, and I know every machine and operator very well, that the plant can run smoothly. Your new system should be able to access different databases on machines and operators and pool the data before it can help me with the timetable. And I guess, once you sort it out, it should be able to help me generate reports for my manager and team as well.

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INTERVIEW 03: Ally Float - Product Engineer at FloatFry

Q	<i>Hi. What have you been busy with lately?</i>
A	Hi. Yeah, I guess you came at the right time. I mean, we have been having this plan for a new product release. Well, it is still within the product lines we currently have, but the new product will have new features and, most importantly, present a totally different character. So, we currently have four product lines of customizable cooking sets, the Cardamon, Chive TX5, Saffron XP, and the Clove TM47. Within those lines, our customers can customize the colour, size, and lid and handle materials, and they can even put their signature on them. And now, our new upcoming product, Rosemary TS1, will feature a new design, new materials, and new colourways. Totally different character and a different feel. It is going to be exciting for us and our customers.
Q	<i>And what is your involvement in this?</i>
A	I am the principal Product Engineer for this product. I have been leading a team of product engineers, or Prodees as they call us, to perform research and development of the product, as well as to work on the design of the product, which includes generating product design specifications. We design the product, choose the materials, quality check the product before release, and basically drive the production. Well, we don't make the final decision, that is for the Production Manager, and we don't run the machines, that one is the Floor Manager's responsibility. But we definitely are an important part of the production and manufacturing processes.
Q	<i>You mentioned driving the production, what are your typical daily activities?</i>
A	We, the Prodees, proudly have and walk our own design and engineering processes called product lifecycle, which spans from conceptualizing and designing to selecting materials. I expect the Manufacturing Resources Planning you are developing to cover all aspects of this. Annabel, our Production Manager, will give you details about our production lines and business processes if you have a chance to see her.

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	Now, say we have done the design process, the main deliverables are [1] the detailed technical drawing, which defines the shape and dimensions of all the components, parts, and pieces that go into the product; [2] the complete set of design specifications, which includes the materials to be used; and [3] the manufacturing instruction, which explains how the materials should be treated, what processes are involved and the sequence, and how the components should be put together to make the final product. All these deliverables will go to the Production Manager, who will subsequently share them with the Floor Manager and our POs.
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INTERVIEW 04: Jakub Nowicki - Production Operative at FloatFry

Q	<i>So, what do you expect from the newly proposed Manufacturing Resources Planning?</i>
A	Well, I am expecting quite a lot from it, to be honest. I think it is going to be good for me, the Prods, and the floor team. The Prodees should be able to upload the design specifications to the network. I should be able to analyse the design online and double-check if the design specifications fit the material availability and the capability of the machines. I mean, I can use some automation in checking the logistics of materials and placing purchase orders of materials immediately when needed. I am also responsible for keeping tabs on the manufacturing processes. You know, if they are going smoothly, or if there is any problem. If the system allows me to notify my manager and my team immediately, that would be great.

INTERVIEW 05: Annabel Float - Production Manager at FloatFry

Q	<i>Thank you for your time. Could you please explain to me about your job?</i>
A	No problems. I am the Production Manager here at FloatFry. Just for your information, this is currently the only FloatFry's manufacturing plant. However, we plan to add more plants soon. It should be realized by the third quarter of 2027. But I need to impose that we will be adopting fully our business approach and processes we have here because they are the embodiment of our vision and idealism since we started our factory in 1993.

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	Right, back to my job description. I work together with Abby, the Floor Manager, to make sure the production and manufacturing processes go smoothly. My teams include Prodees and POs. Abby is leading other teams consisting of Stampers. I am responsible for making sure my team produce the required deliverables on time, which are the availability of materials, the technical drawing, the design specifications, and the manufacturing instruction. We will then bring this information to meetings with Abby and her team. The information should make the foundation to define the manufacturing processes, the jobs, what machines are required, how the production line will look, the production schedule, the operator timetable, what materials are required as well as their availability, and so on.
Q	<i>How can the newly proposed system make your and your team's job easier?</i>
A	Now, you need to know that I also work very closely with David and his team to understand the weekly demand and shipping schedule. David Float is the head of the Marketing department responsible for informing us how many sets we need to produce in any particular week and the specificity of the orders. Once I have the details, I start performing stock diagnosis to decide if we need to contact our suppliers. A long time ago, our suppliers were simply local businesses. However, since we increased the production capacity, our local partners have not been able to keep up with the demand, so we had to work with other companies across the country. Consequently, the availability and delivery time of materials are a major concern for FloatFry. Without effective supply chain management, our production will not run smoothly, or may even stop completely. So, for a start, we would be happy if the new system would allow us to check the capability of the machines, what kind of jobs they can do, how long it takes for them to finish a cycle, what the capacity is, <i>et cetera</i>. Just the general information about the machines to help us with the design process and an early estimation of the production costs will be fine. We are never the expert in those manufacturing aspects, Abby's team are. So, we will still need to consult them later on. More importantly, we need to check if the required materials are in stock, and if not, we

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should be able to place an order directly with our suppliers and reschedule the production if necessary. Abby's team will then help with designing the production line and generating the timetable for the manufacturing jobs. At the same time, my team will need to upload the technical drawings, the design specifications, and the manufacturing instructions to the network so that they will be available on the machines involved in the production line. Since Abby's team only deals with the machines and manufacturing processes, our POs will make sure we have all the materials we need. They will sort the materials, prepare them in batches, and dispatch them to the machines, along with the job cards. This way, the Stampers can go straight to manufacturing the product. When a cycle finishes, the Prodees will come down to the plant and check the quality of the batch and report back to me. If there are any problems, we halt the manufacturing process whilst trying to fix the problems. If everything is fine, we go straight to the next batch.

INTERVIEW 06: David Float – Marketing Director at FloatFry

Q <i>What are the reasons behind the initiative to develop new Manufacturing Resources Planning?</i>	
A	Well, we are proud that FloatFry's products have most recently been voted the Best High-End Cookware of 2021. That shows how our hard work for the last couple of decades is not going to waste. We have grown very rapidly and keep doing so at the moment; and the people, our customers, are supporting us. However, at the same time, we understand that this growth does not come cheap. As our business grows, so must our organization, our staff, our infrastructures, our systems, and basically every aspect of the business. If we find any imbalance and don't do anything about it, we may force the business to have a major dip very soon. That is why we have initiated a long-term business process overhaul. It is in the interest of creating sustainability for the business. And the redesigning of the production and manufacturing processes, including the development of new Manufacturing Resources Planning, is part of our long-term vision.

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Q	<i>I can imagine that the business process reengineering you just mentioned is going to be a significant job. How do you envision the new Manufacturing Resources Planning as a part of the job?</i>
A	Well, I have to emphasize that the overhaul is not necessarily reengineering. I mean, we will still be operating based on Just-in-Time approach. We will not mass produce recklessly. Our production will still strictly based on the incoming orders. We will spend as little money as possible on logistics and storages. We are proud of how we do things and would like to continue. However, we would like to start automating and digitizing most of the processes to increase efficiency and effectiveness. At the end, we want our customers to feel special, but we do not want them to wait too long for our products to arrive on their door steps. I have been eyeing integration between digital tools serving the production floor, the Prods, the marketing department, and the executives for some time now. What I want, or should I say what we need, is an integrated system to plan, manage, and monitor all the production and manufacturing processes of our accessories product lines. This is by no means a simple job to do. Firstly, the new system needs to be able to record orders in real-time and estimate short-time sales accurately. Then, it needs to reach out deep to retrieve real-time data from the machines. It needs to be able to communicate with the plant's resources, such as hardware, legacy software, and other computing resources, as well as acquire knowledge resources from the staff, and the production and manufacturing standards. Next, the system needs to be able to pool these real-time data and create virtualization that enables access to the data clusters. In terms of services, the system should have a middle layer to handle user data, task management, timetabling, resource management, <i>et cetera</i>. Finally, the users, including the Prods, Prodees, Stampers, and executives need to be able to create, design, manage, monitor, overview, and get reports from the system.

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